Chapter test

15 Growth and development

Name: _____ ( ) Class: _____ Date: _____

Total marks: 40 marks
Time allowed: 40 minutes
Answer ALL questions.

Score: / 40

A. Concept checking (10 marks, 1 mark each)

1. Which of the following correctly defines the 'development' of an organism?

   A. increase in height or length of the organism
   B. increase in dry mass of the organism
   C. increase in complexity of the organism
   D. increase in cell size of the organism ☐

2. Which of the following are possible parameters for measuring the growth of a child?

   (1) body weight
   (2) hand span
   (3) circumference of head

   A. (1) and (2) only
   B. (1) and (3) only
   C. (2) and (3) only
   D. (1), (2) and (3) ☐

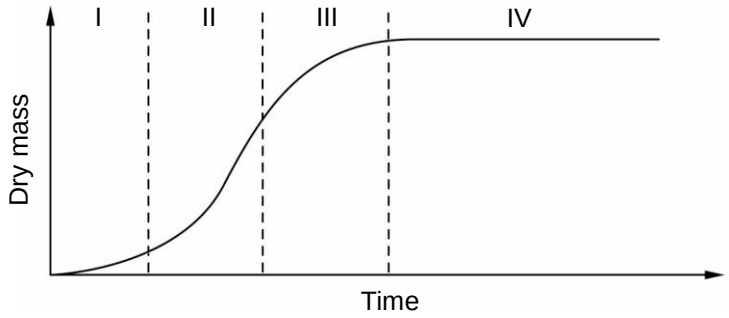
★ 3. Which of the following statements about growth is/are correct?

   (1) The increase in length in plants occurs behind the tips of shoots and roots.
   (2) The increase in the length of stem is mainly due to the increase in cell number.
   (3) Unlike plants, animals continue growing throughout their lives.

   A. (1) only
   B. (1) and (2) only
   C. (2) and (3) only
   D. (1), (2) and (3) ☐

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★ 4. The graph below shows a growth curve of an organism.

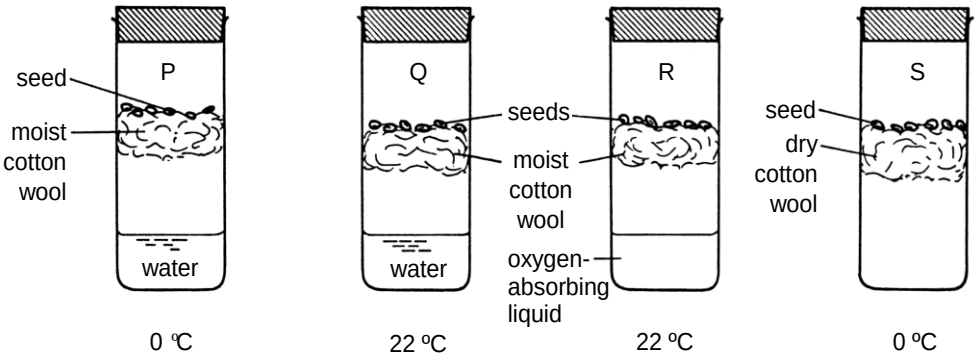


Which of the following statements about the growth curve are correct?

(1) No growth occurs in phase IV.
(2) This is a growth curve of an annual plant.
(3) The growth rate is limited by internal and external factors in phase I and III.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

★ 5. The diagram below shows the experimental set-ups used to investigate the conditions necessary for seed germination.



Which of the following set-ups can be used to investigate the effect of temperature on seed germination?

A. P and Q only
B. P and R only
C. P, Q and S only
D. P, R and S only

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***Directions*:** Questions 6 and 7 refer to the diagrams below. Diagram 1 shows the longitudinal section of a root. Diagram II shows a photomicrograph of some cells taken from the root.

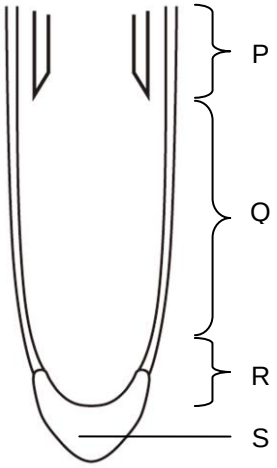


Diagram I

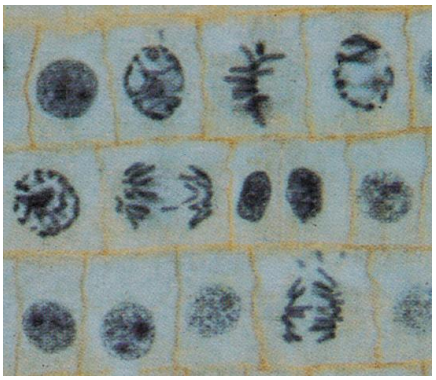


Diagram II

★ 6. The cells shown in Diagram II are most likely taken from region

A. P.
B. Q.
C. R.
D. S. ☐

★ 7. Which of the following is the major cause that accounts for the increase in the length of the root?

A. cell division at region Q
B. cell division at region R
C. cell elongation at region Q
D. cell elongation at region R ☐

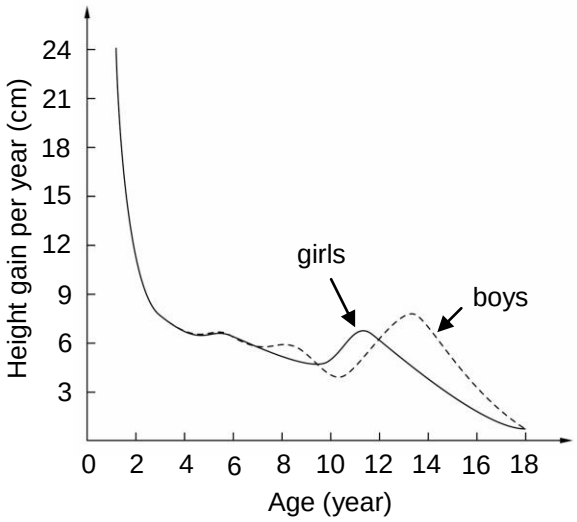
8. Which of the following belong(s) to the secondary growth of a plant?

(1) increase in the length of the stem
(2) increase in the thickness of the root
(3) increase in the number of leaves

A. (2) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only ☐

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★ 9. The graph below shows the height gain per year in boys and girls.



Which of the following can be deduced from the graph?

(1) The height of boys and girls are the same at the age of 12.

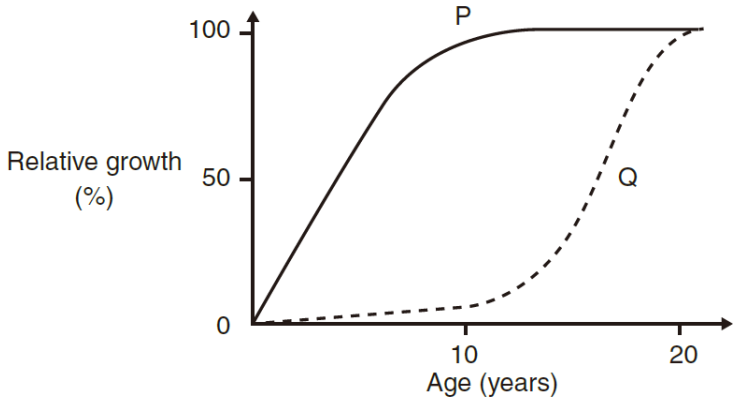
(2) The increased rate of growth in boys is greater than in girls during adolescence.

(3) Boys secrete more sex hormones than girls during adolescence.

A. (2) only B. (1) and (2) only

C. (1) and (3) only D. (2) and (3) only

10. The graph below shows the relative growth rates of two different parts of the human body.



Which of the following combinations correctly identifies curves P and Q?

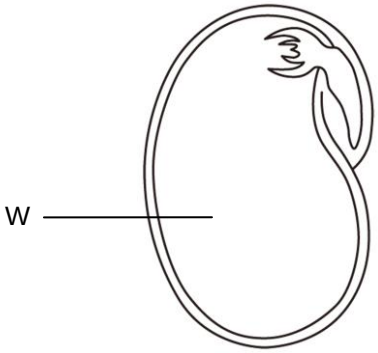
	P	***Q***
A.	brain	whole body
B.	brain	reproductive organs
C.	whole body	brain
D.	whole body	reproductive organs

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Question	1	2	3	4	5	6	7	8	9	10
Related section	15.1	15.2	15.4	15.2	15.3	15.4	15.4	15.4	15.5	15.5

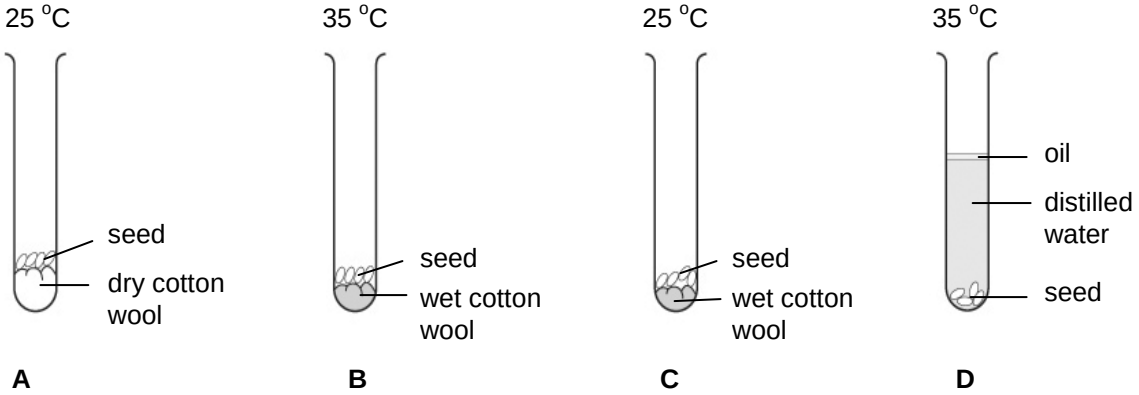
B. Problem solving (22 marks)

★ 11. The diagram below shows a section of a mung bean seed.



(a) Name structure W. During germination, amylase activity can be detected in structure W. Explain the importance of amylase activity in seed germination. (4 marks)

(b) A student carried out an experiment to investigate the condition necessary for seed germination.

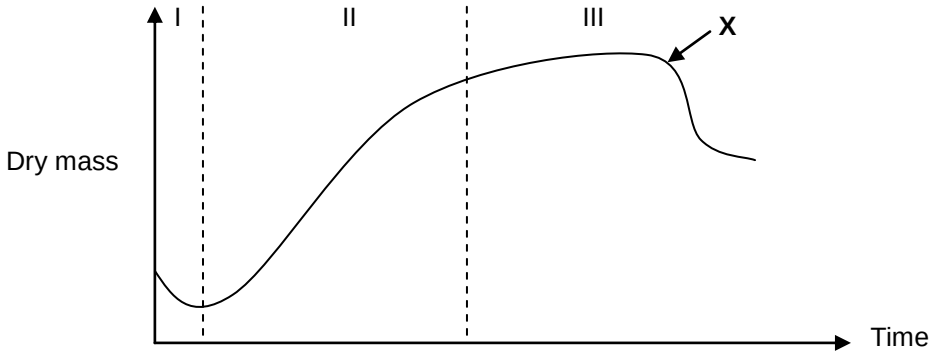


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In which tube would the seeds germinate first? Explain your answer. (4 marks)

(Total: 8 marks)

12. The graph below shows the growth curve of a pea plant.



(a) Briefly describe a way to measure the dry mass of a pea seedling. (2 marks)

(b) Describe and account for the change in dry mass of the plant in phase I. (3 marks)

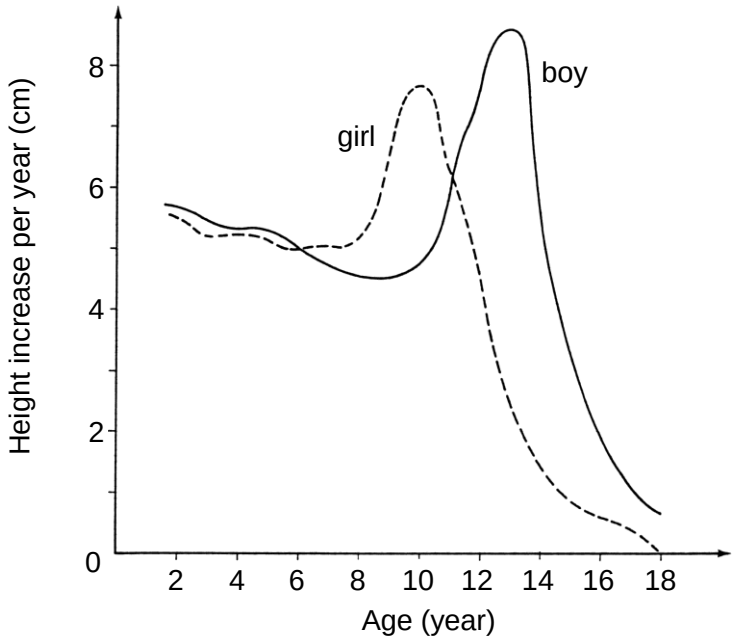
(c) Explain why the dry mass of the plant gradually increases in phase II. (3 marks)

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(d) Suggest why there is a sudden decrease in dry mass at point X. (1 mark)

(Total: 9 marks)

★ 13. The graph below shows the height increase of a boy and a girl from age 1 to age 18. The data were recorded in each year.



(a) At which age(s) do the boy and girl have the same height increase? (1 mark)

(b) According to the graph, state ***two*** differences between the adolescent growth patterns of the boy and girl. (2 marks)

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(c) Apart from the increase in body height and weight, state ***two*** physical changes in the boy at the age of 13. (2 marks)

(Total: 5 marks)

C. Exam challenge (8 marks)

★★14. The photograph below shows the section of a ripe tomato fruit.



(a) John observed that tomato seeds do not germinate inside tomato fruits. He suggested that there are substances in tomato fruits that stop the seeds from germinating while they are inside the tomato fruits. Design an experiment to investigate whether tomato fruits contain substances that inhibit the germination of tomato seeds. (5 marks)

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(b) Explain why fresh mass and dry mass are not suitable parameters for measuring the growth of a germinating tomato seedling over a period of time. Suggest a suitable parameter for the measurement. (3 marks)

(Total: 8 marks)

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Marking scheme

A. Concept checking (10 marks, 1 mark each)

1.	C	2.	D	3.	A	4.	B	5.	A
6.	C	7.	C	8.	A	9.	A	10.	B

(1 mark each)

B. Problem solving (22 marks)

11. (a) Cotyledon (1)
Amylase hydrolyses the starch stored in structure W into maltose / sugars (1)
which can be used in respiration to provide energy for growth (1)
and used for forming new cells / structural components of new cells for growth. (1)

(b) Tube B (1)
Tube A does not contain water. Enzymes inside the seeds are not activated to break down the food reserves for growth. Thus, the seeds in tube A cannot germinate. (1)
The seeds in tube D are covered by oil. There is insufficient dissolved oxygen available for respiration of the seeds. Thus, the seeds in tube D cannot germinate. (1)
The temperature of tube B is higher than that of tube C. Warmer temperature allows enzymes to work more efficiently. Thus, the seeds in tube B will germinate earlier. (1)

(Total: 8 marks)

12. (a) Dry the seedling in an oven at a temperature slightly above 100 ^{o}C (1)
until a constant mass is obtained upon repeated weighing. (1)

(b) The dry mass decreases in phase I. (1)
It is because the food reserves in the seed are broken down (1)
to provide energy for growth. (1)

(c) Green leaves are developed and start to carry out photosynthesis. (1)
When the rate of photosynthesis is higher than the rate of respiration, (1)
there is a net gain in the amount of organic matter in the seedling. (1)

(d) The decrease in dry mass at point X is due to the dispersal of seeds and fruits. (1)

(Total: 9 marks)

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13. (a) At the age of 6 and 11 (1)

(b) The growth spurt (or period of rapid growth) of the boy occurs later than that of the girl. (1)

The increased rate of growth (or maximal height increase per year) in the boy is greater than in the girl. (1)

(c) Any ***two*** of the following: (2)

- Growth of beard and pubic hair (1)
- Enlargement of larynx (1)
- Broadening of shoulders (1)
- Increase in muscle mass (1)

(Or other reasonable answers)

(Total: 5 marks)

C. Exam challenge (8 marks)

14. (a) Remove seeds from tomatoes and crush the tomatoes to make tomato juice. (1)

Experimental set-up: Put a piece of cotton wool to a Petri dish. Add a fixed volume of tomato juice to the dish. Put a sufficiently large number of tomato seeds in the dish. (1)

Control set-up: same set-up with using distilled water instead of tomato juice (1)

Leave both set-ups in a warm and well-ventilated place for the same period of time. (1)

Count and compare the number of germinated seeds in each dish to see if there is any significant difference. (1)

(b) Measuring dry mass: The seedling is killed during the measurement. It is impossible to make a continuous measurement on it.

/ A large number of seedlings are needed in order to measure the average growth over a period of time. (1)

Measuring fresh mass: The fresh mass of the seedling is easily affected by factors unrelated to growth (e.g. water content of the soil). (1)

Measure the length of the shoot / total surface area of the leaves of the seedling. (1)

(Total: 8 marks)

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